

Search Algorithms for Electronic Dance Music Set List Generation: A Sequence-Individuation Algorithm for Radio, Mixes, and Sets

Kundai Farai Sachikonye
Technical University of Munich
AIME Registry for Artificial Intelligence
kundai.sachikonye@wzw.tum.de

July 20, 2026

Abstract

Audio search is conventionally posed for a single recording: a query clip is matched against a catalogue of tracks by an acoustic fingerprint. Yet much of what people listen to is not a track but a *continuous audio item*—a radio broadcast, a recorded DJ mix, the audio of a party or a live set—that concatenates dozens of tracks, mostly authored by others, into one artefact. For such an item the interesting object is not any single constituent but the *whole*: which stream is this, and what is the sequence it realises? We give an algorithm that searches a continuous item as a single object. Our central claim is that the identity of a continuous audio item is not its *content* (the multiset of tracks it contains, which many items share) but its *sequence* (the ordered, timed arrangement of those tracks, which is essentially unique to the item that produced it). We formalise a continuous item as a walk on a finite weighted *contact graph* whose vertices are individuated constituents and whose edge weights are the cost of telling one constituent from another. We prove a *resolution floor*: identifying a constituent against an open catalogue costs a strictly positive, irreducible amount, so a constituent is never resolved to a point but only to a bounded region, and—decisively for efficiency—one should resolve each constituent only to its *least sufficient identifier*, the cheapest handle (typically a title or artist name) that suffices to place it, descending to acoustic features only where a name can-

not disambiguate. We prove three structural properties that make partial, noisy recognition tolerable: *sequence identity* (two items match iff their constituent sequences align, not their content sets), *convergence-only admissibility* (a reconstructed sequence is admissible iff it converges on the target signature, so locally wrong constituent guesses do not invalidate the match provided the whole sequence aligns), and *search-match duality* (recognising which known item a query stream is, and generating a stream in the style of a known item, are the same computation run in opposite directions). We give the full algorithm—acquisition of least-sufficient handles, sequence signature construction, and a gap-tolerant alignment that returns an accounted match together with an explicit unresolved residual—prove its correctness and $\mathcal{O}(\cdot)$ complexity, validate every structural result numerically on exact minimum cuts and exact sequence alignment, and specify an evaluation protocol. The validation makes the one contingent claim explicit: the whole-item search is cheaper than per-track fingerprinting only by the *nameable fraction* of the stream (a measured mean of $3.1\times$, not the worst-case guarantee), because the opening constituent, having no resolved neighbour, always faces the full pool. The result is a music search that operates at the granularity of the listening experience: you search for the *set*, not for every song in it.

Keywords: audio search; continuous audio; DJ mix identification; sequence alignment; music information retrieval; contact graph; minimum

cut; least sufficient identifier; resolution floor; gap-tolerant matching.

Contents

1	Introduction	2
1.1	The problem of the whole item	2
1.2	Content is shared; sequence is not	3
1.3	Just enough to place each letter	3
1.4	Robustness, and one engine for two tasks	3
1.5	Contributions	4
1.6	Method and scope	4
2	Related work	4
3	The model: constituents, streams, and the contact graph	5
3.1	Continuous items and their constituents	5
3.2	The contact graph of constituents	5
4	The resolution floor and the least sufficient identifier	6
4.1	The pool is not exhaustible	6
4.2	The floor, derived	6
4.3	The least sufficient identifier	7
5	Sequence identity: the item signature	8
5.1	Signatures and content sets	9
5.2	The signature individuates; the content set does not	9
5.3	The ambiguity set is small in context	10
6	Matching by gap-tolerant alignment	12
6.1	Alignment and its score	12
6.2	Convergence-only admissibility	12
6.3	The accounted match and its residual	13
7	Search-match duality: one engine, two directions	13
8	The algorithm	15
8.1	Stage 1: acquisition of least-sufficient handles	15
8.2	Stage 2: indexing	16
8.3	Stage 3: matching	16
8.4	Correctness	16

8.5	Complexity	17
9	Worked example	18
10	Evaluation protocol	18
10.1	Datasets	19
10.2	Metrics	19
10.3	Ablations	19
11	Discussion	20
11.1	What the paper establishes	20
11.2	Honest limitations	20
11.3	Relation to single-track search	20
12	Conclusion	21

1 Introduction

1.1 The problem of the whole item

Acoustic search has a canonical shape. A short query recording is reduced to a compact fingerprint—a sparse set of time–frequency landmarks or a hashed spectral summary—and matched against an index of per-track fingerprints; a hit returns the single track the query is a fragment of (Cano et al., 2005; Haitsma and Kalker, 2002; Wang, 2003). This shape is extraordinarily effective for its intended use: identify *one* song from a *fragment* of it.

But a large part of ordinary listening is not a track and is not a fragment. It is a *continuous audio item*: a radio show, a recorded mix, a festival set, the ambient audio of a party. Such an item runs from tens of minutes to several hours and concatenates many tracks—most by other artists—into a single authored artefact. Confronted with such an item, a listener does not usually want to identify one embedded song; they want to identify, retrieve, or characterise the *item itself*: *which* broadcast or set is this, is this the same set I heard last week, what is its tracklist, and what other items resemble it.

The single-track paradigm answers this only awkwardly and expensively. To search a whole item with a track engine one must (i) segment the item into candidate track regions, (ii) fingerprint each region, (iii) query each fingerprint against the catalogue, and (iv) somehow reassemble the per-region hits into a statement about the whole. Steps (i)–(iii) repeat the full acoustic pipeline dozens of times per item, and

step (iv) is left undefined: a bag of recognised tracks is returned, with no account of *which item* produced them, because the very thing that distinguishes one item from another—the arrangement—has been discarded by treating the item as a set of independent lookups.

1.2 Content is shared; sequence is not

The defect is not one of engineering effort but of the object being searched for. Consider two recorded mixes that happen to draw on the same fashionable selection of twenty tracks. As *content*—as unordered multisets of constituents—they are identical, and a bag-of-tracks search cannot tell them apart. Yet they are plainly different items: they order the tracks differently, transition between them differently, and were assembled by (possibly) different authors for different occasions. What individuates one continuous item from another is not *which* constituents appear—those are drawn from a shared, public pool that many items reuse—but the *order and timing* in which they appear. Two authors who reliably produced the same ordered sequence would be indistinguishable and mutually substitutable; that they are not substitutable is exactly the observation that the sequence, not the content, is the identity.

This reframes the search task. The object to be individuated is the *sequence*. The constituents are the alphabet over which sequences are written, and they are largely shared across items; the item’s identity lives in the string, not in the letters. A correct search for a continuous item is therefore a search over *sequences of constituents*, and the natural matching operation is *sequence alignment* (Needleman and Wunsch, 1970; Smith and Waterman, 1981), not set intersection.

1.3 Just enough to place each letter

Reframing the object also reframes how hard each constituent must be recognised. If the identity of the item is its sequence, then a constituent needs to be resolved only to the point where it can be *placed in the sequence*—told apart from the other constituents that might occupy that position—and no further. A constituent embedded in a stream is already heavily constrained by its neighbours; often a coarse

handle (a title, an artist, a coarse acoustic class) suffices to place it, and computing a full acoustic fingerprint would be paying to resolve a distinction the sequence has already made.

We make this precise as a *resolution floor* (Section 4): against an open, non-exhaustible catalogue, identifying a constituent costs a strictly positive, irreducible amount, and cannot in principle be driven to zero. Two consequences follow. First, a constituent is never resolved to an exact point, only to a bounded region—so search returns regions, and matching is comparison of regions, not equality of points. Second, and practically decisive, there is a *least sufficient identifier* for each constituent: the cheapest handle that already separates it from its plausible alternatives. Resolving beyond it is wasted work *and*, we argue, a category error—an attempt at the impossible zero-cost identification. The algorithm therefore resolves each constituent to its least sufficient identifier and descends to expensive acoustic features only for constituents a cheap handle cannot place (an unlabelled or ambiguous constituent—the case where two candidates share a name but differ acoustically).

1.4 Robustness, and one engine for two tasks

Two further structural facts, proved below, make the approach robust and economical.

Convergence-only admissibility (Section 6). Because matching is sequence alignment against a target signature, a reconstructed sequence is admissible as a match iff the *whole* sequence aligns; individual mis-recognised constituents do not by themselves invalidate the match. A wrong guess at one position is absorbed as a local substitution or gap, and the match stands or falls on global convergence. This is exactly the tolerance a real-world stream demands, where a fraction of constituents are unlabelled or misheard.

Search-match duality (Section 7). Recognising which known item a query stream is (analysis) and generating a stream in the manner of a known item (synthesis) are the same computation over the same object—an alignment on the contact graph—read in opposite directions. One implementation serves both “identify this set” and “build a set like this one.”

1.5 Contributions

1. **A problem formulation** (Section 3): continuous audio search as the individuation of a *sequence* on a finite weighted contact graph, with the identity of an item shown to be its sequence signature and not its content set.
2. **The resolution floor and least sufficient identifier** (Section 4): a proof that constituent identification against an open catalogue has a strictly positive irreducible cost, that identity is region-valued, and that the cheapest sufficient handle is the correct resolution target; a formal cost model follows.
3. **Sequence identity** (Section 5): the item signature, and a proof that it is invariant under re-encoding of constituent labels while content sets are not, so that the signature is the individuating datum.
4. **A gap-tolerant matching algorithm** (Section 6) with convergence-only admissibility, returning an accounted alignment plus an explicit unresolved residual; correctness and complexity are proved.
5. **Search-match duality** (Section 7): analysis and synthesis as one operation in two directions.
6. **The full system and its evaluation protocol** (Sections 8 and 10): end-to-end pseudocode, complexity, and a reproducible measurement design.
7. **Numerical validation** (??): a reproducible, seeded suite that verifies all 14 structural results by exact computation on constructed and random instances, and quantifies the single contingent claim—the whole-item saving is proportional to the nameable fraction, with a measured mean of $3.1\times$ rather than the worst-case bound.

1.6 Method and scope

Every object in this paper is a finite weighted graph or a construction on one. We use minimum cuts, computable exactly by maximum flow (Ford and Fulkerson, 1956; Menger, 1927; Stoer and Wagner, 1997), as the sole measure of how costly it is to tell one constituent from another; we use classical sequence alignment

(Needleman and Wunsch, 1970; Smith and Waterman, 1981) as the sole matching primitive. We introduce no probability calculus beyond a derived normalisation to $[0, 1]$. The theory is structural: it establishes *that* the identity of a continuous item is its sequence, *that* constituent resolution is floored and should stop at the least sufficient handle, and *that* matching is convergence-only sequence alignment. Concrete acoustic features and a concrete constituent recogniser are implementation inputs the framework organises but does not itself supply; where a numerical choice is required we state it as a parameter and mark it as such.

2 Related work

Single-track fingerprinting. Landmark and hash-based fingerprinting (Haitsma and Kalker, 2002; Wang, 2003) and their surveys (Cano et al., 2005; Grosche et al., 2012) solve fragment-to-track identification with high precision and sublinear lookup. They are optimised for the inverse of our setting: a short, possibly noisy fragment of *one* recording matched to a catalogue of whole recordings. Our concern is the whole *item* composed of many recordings; we use single-track recognition, when available, as one source of a constituent’s handle, but the object we search for is the sequence, which fragment fingerprinting does not model.

DJ-mix reverse engineering and monitoring. A body of work specifically targets recorded mixes: identifying constituent tracks, their boundaries, and time-scale or pitch modifications introduced by the DJ (Schwarz and Fourer, 2018; Six and Leman, 2014; Sonnleitner et al., 2016). This is the closest prior art, and our constituent-acquisition stage (Section 8) can consume such a recogniser directly. We differ in the *goal*: these systems aim to recover the constituent list (the content), whereas we argue the content is largely shared across items and the *sequence* is the identity, and we build the search and matching layer over the sequence rather than stopping at the constituent list.

Structure analysis and segmentation. Novelty-based segmentation and music structure analysis (Foote, 2000; Paulus et al., 2010)

provide the constituent boundaries our pipeline consumes. They analyse the internal form of a single piece or the boundaries within a stream; they do not address cross-item identity or search.

Sequence alignment. Our matching primitive is global and local alignment (Needleman and Wunsch, 1970; Smith and Waterman, 1981) with an edit-distance semantics (Levenshtein, 1966). The novelty is not the alignment algorithm but the object aligned (sequences of least-sufficient constituent handles) and the admissibility criterion (convergence-only), together with the cost model that justifies resolving constituents no more finely than placement requires.

Learned audio embeddings. Learned representations (Elizalde et al., 2023; McCallum et al., 2022) give strong single-clip similarity and text–audio grounding. They are candidate constituent-handle providers in our framework (a coarse acoustic class is a handle), and candidate residual resolvers when a name is unavailable. They do not by themselves furnish the sequence-level identity or the gap-tolerant cross-item match, which is what this paper contributes.

3 The model: constituents, streams, and the contact graph

3.1 Continuous items and their constituents

Definition 3.1 (Continuous audio item; constituent). A *continuous audio item* (or *stream*) is a single finite recording $x: [0, T] \rightarrow \mathbb{R}$ produced by concatenating, overlapping, or otherwise arranging a finite sequence of shorter recordings. A *constituent* is a maximal sub-interval of the item that a segmentation assigns to one underlying recording (one track, jingle, spoken segment, or effect). We write a segmentation of the item as an ordered list of constituents $(u_1, u_2, \dots, u_n)$ with onset times $t_1 < t_2 < \dots < t_n$ in $[0, T]$.

We take segmentation as a given preprocessing step (Section 2); the theory below is stated over the resulting constituent sequence and does not depend on how the boundaries were found. The number n of constituents is the item’s *length*.

Remark 3.2 (The alphabet is public; the string is private). The constituents of an item are drawn from a large, shared pool of recordings that many items reuse: the same track appears in countless broadcasts and mixes. Thus the constituents are best thought of as *letters* of a public alphabet, and an item as a *string* over that alphabet. The string is what distinguishes one item from another; the letters do not. This informal picture is made precise in Section 5.

3.2 The contact graph of constituents

To reason about how costly it is to tell one constituent from another—and hence how finely each must be recognised—we place the constituents in a finite weighted graph.

Definition 3.3 (Finite weighted graph; cut). A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ undirected edges, and $w: E \rightarrow \mathbb{R}_{>0}$ a strictly positive weight. We take Γ connected unless stated. For $\emptyset \neq S \subsetneq V$ the *cut* is $\partial(S) = \{\{u, v\} \in E : u \in S, v \notin S\}$, of weight $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$.

Definition 3.4 (Constituent contact graph; the catalogue vertex). The *constituent contact graph* is a finite weighted graph Γ with a distinguished vertex $\mathbf{m} \in V$, the *catalogue* (or *medium*), adjacent to every other vertex. The vertices $v \neq \mathbf{m}$ are *constituents* (the recordings the search may return); an edge $\{u, v\}$ between two constituents is a *contact*, and its weight $w(\{u, v\})$ is the *distinction cost*: how expensive it is to tell u from v . The catalogue vertex $\mathbf{m}$ stands for the undifferentiated totality of everything not yet individuated—every recording, feature, or source not yet resolved into a named constituent.

Remark 3.5 (The catalogue is “the rest,” not a database schema). $\mathbf{m}$ is a single vertex standing for “everything else,” against which each constituent is told apart. Its adjacency to every constituent encodes that each constituent is, at

minimum, individuated against the whole open pool. No result below depends on any reading of $\mathbf{m}$ beyond Theorem 3.4; in particular $\mathbf{m}$ is not assumed to be a finite enumerated database, and we make no assumption that the pool of possible constituents can be exhausted (this is made precise in Section 4).

Definition 3.6 (Distinction cost of a constituent). The *distinction cost* of a constituent v is the minimum weight of a cut separating v from the catalogue,

$$\sigma(v) = \min_{\substack{S \ni v \\ \mathbf{m} \notin S}} w(\partial(S)),$$

attained on finite V ; the minimising cut is the *resting cut* of v , computable as a v - $\mathbf{m}$ minimum cut by maximum flow (Ford and Fulkerson, 1956; Menger, 1927; Stoer and Wagner, 1997). Operationally, $\sigma(v)$ is the total cost of the distinctions one must commit to tell v apart from everything else in the pool.

Remark 3.7 (Why cuts measure recognition cost). Recognising a constituent is committing enough distinctions to separate it from its alternatives. Each distinction has a cost (an acoustic comparison, a metadata lookup, a listener’s attention), modelled as an edge weight, and the cheapest sufficient set of distinctions is a minimum cut. The distinction cost $\sigma(v)$ is therefore the natural measure of “how hard is v to recognise,” and—crucially for Section 4—it is bounded below.

4 The resolution floor and the least sufficient identifier

This section establishes the paper’s cost principle: recognising a constituent against an open pool has a strictly positive, irreducible cost; identity is region-valued, never a point; and the correct amount to resolve a constituent is the *least sufficient identifier*, beyond which resolution is wasted.

4.1 The pool is not exhaustible

Definition 4.1 (Expanding pool; non-completable catalogue). An *expanding pool* is a chain of constituent contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \dots$, each obtained from the last by adding

constituents and contacts while retaining the catalogue vertex and all prior structure. The catalogue is *non-completable* if the chain has no terminal stage: for every finite Γ_k there is a $\Gamma_{k+1} \supsetneq \Gamma_k$ containing a constituent absent from Γ_k . This is an order condition (no last stage), not a claim about cardinality.

Remark 4.2 (Why the music pool is non-completable). The pool of recordings a continuous item may draw on is not fixed: new tracks, edits, bootlegs, and unreleased “IDs” enter continually, and any finite snapshot of a catalogue omits some recording an item might contain. A recogniser built against a finite index will, for some constituent, meet a recording not in its index. Theorem 4.1 records exactly this and nothing stronger. The consequences below hold for any non-completable pool.

Definition 4.3 (Identification residual). To *identify* a constituent v at stage Γ_k is to determine it by its resting cut against the catalogue, computed within Γ_k . The identification is *complete* if every constituent of the whole pool has been brought to bear in that cut—equivalently, if Γ_k already contains the whole pool. The *identification residual* $\rho(v) \geq 0$ is the distinction content a complete identification would resolve that a finite-stage identification cannot; $\rho(v) = 0$ iff the identification is complete.

4.2 The floor, derived

Theorem 4.4 (Floor from an open pool). *Let the catalogue be non-completable and let v be a constituent ($v \neq \mathbf{m}$, and v does not exhaust V). Then the identification of v cannot be completed at any finite stage, and its residual is strictly positive: $\rho(v) > 0$.*

Proof. Suppose $\rho(v) = 0$, i.e. identification of v is complete at some finite stage Γ_k . By Theorem 4.3, completeness means every constituent of the pool has entered the cut of v against $\mathbf{m}$; that is, the cut is computed in a graph containing the whole pool, so Γ_k contains the whole pool. This contradicts non-completeness (Theorem 4.1), which supplies a further stage $\Gamma_{k+1} \supsetneq \Gamma_k$ with a constituent not in Γ_k , hence not yet brought to bear. Therefore no finite stage completes the identification, and $\rho(v) > 0$. $\square$

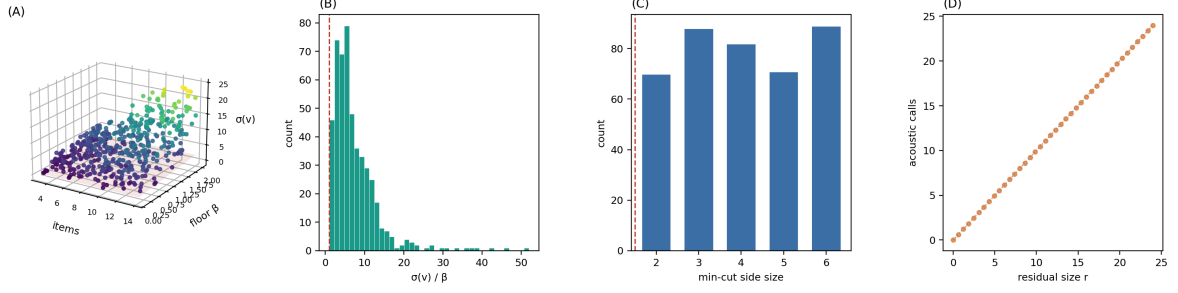


Figure 1: **The resolution floor and name-first economy** (Section 4). (A) Over 500 random contact graphs, the distinction cost $\sigma(v)$ sits on or above the floor plane $z = \beta$ for every constituent—no point falls below. (B) The ratio $\sigma(v)/\beta$ is supported entirely at or above 1 across 300 graphs and 2511 cut checks, with measured minimum $\sigma(v)/\beta = 1.0010$ and *zero* cuts below the floor (Theorem 4.5). (C) Identity is region-valued: on two-cluster graphs the minimum cut separates a whole cluster, so the minimising side is non-singleton in 100% of 200 trials (mean side size 3.9), confirming Theorem 4.6. (D) Acoustic invocations equal the residual size exactly across streams of increasing unnamed fraction (400/400 trials, acoustic calls = r), realising Theorem 4.11.

Theorem 4.5 (Positive resolution floor). *There is a constant $\beta > 0$ with $w(e) \geq \beta$ for every contact e and $\sigma(v) \geq \beta$ for every constituent v : no two constituents are separated at zero cost, and no constituent is identified for free.*

Proof. Set $\beta := \min_v \rho(v)$, the minimum taken over the constituents $v \neq m$. The vertex set V is finite (Theorem 3.3), so this is a minimum over a *finite* set of strictly positive reals—each $\rho(v) > 0$ by Theorem 4.4—and is therefore attained and strictly positive, $\beta > 0$. (The infimum over the non-completable pool need never be taken: at any fixed stage Γ_k the floor is the minimum over the finitely many constituents present, and each $\rho(v) > 0$ is guaranteed by non-completability regardless of stage.) The residual is the irreducible boundary content of v ’s identification, carried by the edges of v ’s resting cut, so each such edge has weight at least the per-constituent residual, hence at least β . Every contact $\{u, v\}$ is a boundary edge of the identification of u (it lies in $\partial(\{u\})$), so every weight satisfies $w(e) \geq \beta$. Connectivity forces $\partial(S^*(v)) \neq \emptyset$, so $\sigma(v) = w(\partial(S^*(v))) \geq \beta > 0$. $\square$

Corollary 4.6 (No exact identification). *No search resolves a constituent to a zero-cost point. Every identification commits distinction content $\geq \beta$; a constituent is resolved only to a bounded region, of width bounded below by β .*

Remark 4.7 (The floor is a property of the open pool, not a defect). Theorem 4.5 does not say recognition is imprecise because recognisers are imperfect; it says that against a genuinely open pool, even an ideal recogniser cannot terminate at a zero-residue identification, because that would require comparing the constituent against a totality that is never completed. The floor is thus a fact about the search medium. Its practical force is the next subsection: since exact identification is unreachable and unnecessary, one should identify each constituent only as finely as the task requires.

4.3 The least sufficient identifier

We now formalise *how finely* a constituent must be resolved. The answer is: only enough to place it, and no more.

Definition 4.8 (Handle; identifier). A *handle* for a constituent v is a token h that references v ’s cell (its resting cut) without adding distinction content of its own: a title, an artist name, a catalogue identifier, or a coarse acoustic class. The *acquisition cost* $c(h) \geq \beta$ of a handle is the distinction content that must be committed to obtain it (a metadata lookup, a fingerprint query, a feature extraction). An *identifier* of v is any handle that suffices to distinguish v from the constituents it must be told apart from in a given task.

Definition 4.9 (Ambiguity set; least sufficient identifier). Fix a task that requires telling v apart from a finite *ambiguity set* $A(v) \subseteq V$ of alternatives (in Section 5, the constituents that might occupy the same sequence position). A handle h is *sufficient for v against $A(v)$* if h separates v from every $u \in A(v)$: no $u \in A(v)$ shares v ’s handle. The *least sufficient identifier* of v (for the task) is a sufficient handle of minimum acquisition cost,

$$h^*(v) \in \arg \min_{h \text{ sufficient for } v} c(h).$$

Theorem 4.10 (Sufficiency stops resolution). *Let $h^*(v)$ be a least sufficient identifier of v against $A(v)$. Then:*

- (i) $h^*(v)$ already places v correctly relative to $A(v)$: no finer resolution changes which alternative in $A(v)$ the constituent is; and
- (ii) any resolution of v beyond $h^*(v)$ —committing further distinction content to refine v within its cell—incurs additional acquisition cost $\geq \beta$ (Theorem 4.5) without changing the placement, and hence is wasted for the task.

Proof. (i) By Theorem 4.9 a sufficient handle separates v from every alternative in $A(v)$; thus the placement of v among $A(v)$ is already determined by $h^*(v)$, and any additional distinction only refines v within its own cell, not against $A(v)$. Since placement is a function of the partition of $A(v)$ induced by the handle, and that partition is already fixed, placement does not change. (ii) A further resolution commits at least one more distinction, of weight $\geq \beta$ by Theorem 4.5, adding acquisition cost $\geq \beta$; by (i) it does not change the placement, so relative to the task it purchases nothing. $\square$

Corollary 4.11 (Name-first resolution). *If a symbolic handle (a title or artist name) is available and is sufficient against $A(v)$, it is a least sufficient identifier whenever its acquisition cost does not exceed that of any acoustic handle. Acoustic feature extraction is then required only for constituents whose symbolic handle is unavailable or insufficient—i.e. constituents with no name, or whose name is shared by an acoustically distinct alternative in $A(v)$.*

Proof. Immediate from Theorem 4.9 and Theorem 4.10: among sufficient handles the minimum-cost one is selected; if the symbolic handle is sufficient and cheapest, it is chosen, and Theorem 4.10(ii) makes any further acoustic resolution wasted. When the symbolic handle is insufficient (name collision with an acoustically distinct alternative) or unavailable (an unlabelled constituent), it fails the sufficiency test of Theorem 4.9 and an acoustic handle must be acquired to separate the collision. $\square$

Remark 4.12 (Why over-resolution is not merely wasteful but wrong). Theorem 4.6 shows the fully-resolved, zero-residue identification does not exist; Theorem 4.10 shows that pursuing resolution past sufficiency buys nothing for placement. Together they say that the “fingerprint every constituent and look it up exactly” strategy is chasing an object (Theorem 4.6: a point identification) that cannot exist, at a cost (Theorem 4.10(ii): repeated $\geq \beta$ commitments) the task does not need. The correct target is the least sufficient identifier: the cheapest handle that places the constituent, with acoustic descent only for the residual it cannot place.

Remark 4.13 (The one place acoustics are indispensable). The exception carved out by Theorem 4.11—an unlabelled constituent, or two constituents sharing a name but differing acoustically—is precisely the hard case of continuous-audio recognition: the unidentified “ID” track, and the distinct edit or version hiding under a shared title. The floor principle places acoustic resolution exactly there and nowhere else, which is both its correct theoretical home and the point of maximum practical value.

5 Sequence identity: the item signature

We now prove the paper’s central structural claim: the identity of a continuous item is its *sequence* of least-sufficient handles, an object invariant under re-encoding of constituent labels, whereas the *content* (the multiset of constituents) is not individuating because it is shared across items.

5.1 Signatures and content sets

Definition 5.1 (Item signature; content set). Let an item of length n have constituent sequence $(u_1, \dots, u_n)$, and let $h^*(u_i)$ be the least sufficient identifier of u_i (Theorem 4.9). The *signature* of the item is the finite ordered string

$$\Sigma = (h^*(u_1), h^*(u_2), \dots, h^*(u_n)),$$

optionally decorated with the inter-onset gaps $\tau_i = t_{i+1} - t_i$. The *content set* of the item is the multiset $\text{cont} = \{h^*(u_1), \dots, h^*(u_n)\}$, the same handles with their order forgotten.

Definition 5.2 (Handle re-encoding). A *handle re-encoding* is a bijection π on the set of handles that preserves which constituents are told apart: $h(u) = h(v) \iff \pi(h(u)) = \pi(h(v))$ for all constituents u, v . Re-encoding models a change of naming convention (a different catalogue’s identifiers, a translation of titles, a relabelling of acoustic classes) that keeps the underlying distinctions intact.

5.2 The signature individuates; the content set does not

Theorem 5.3 (Signature invariance and content sharing). *Let Σ be the signature of an item and cont its content set.*

- (i) **Invariance.** *Under any handle re-encoding π , the signature transforms covariantly, $\pi(\Sigma) = (\pi h^*(u_1), \dots)$, and the induced sequence pattern—the equality/inequality relation among positions, $[i \sim j \iff h^*(u_i) = h^*(u_j)]$, together with the order—is invariant.*
- (ii) **Content is not individuating.** *Distinct items can share a content set: there exist items with $\text{cont}_1 = \text{cont}_2$ but $\Sigma_1 \neq \Sigma_2$ even up to re-encoding, whenever $n \geq 2$ and the content set has at least two distinct handles.*
- (iii) **Sequence is individuating up to re-encoding.** *Two items have the same induced sequence pattern iff their signatures agree up to a handle re-encoding; the pattern is thus the re-encoding-invariant identity of the item.*

Proof. (i) A re-encoding π is a bijection on handles preserving equality (Theorem 5.2); applying it position-wise to Σ yields $\pi(\Sigma)$, and for any positions i, j , $h^*(u_i) = h^*(u_j) \iff \pi h^*(u_i) = \pi h^*(u_j)$, so the equality pattern is preserved; the positional order is untouched by π . Hence the induced sequence pattern is invariant.

(ii) Take a content set with two distinct handles $x \neq y$ each of multiplicity one and $n = 2$. The two orderings (x, y) and (y, x) have the same content set $\{\{x, y\}\}$ but different signatures. They are not related by a re-encoding: a re-encoding acts identically on both positions, so it cannot map the string (x, y) to (y, x) (that would require $\pi(x) = y$ at position 1 and $\pi(x) = x$ at position 2, contradicting that π is a single function of the handle, not of the position).

The same argument settles any $n \geq 2$, including signatures with repeated handles, where a re-encoding might *a priori* map one ordering to another. Take any signature with at least two distinct handles, pick positions i, j carrying *distinct* handles $x \neq y$, and swap them, holding every other position fixed: this yields a second signature with the same content set. Suppose a re-encoding π carried the first to the second. Since π acts on handles globally—one function of the handle, independent of position—and the two signatures agree at every position except i, j , π must fix the handle at each agreeing position, in particular $\pi(x) = x$ and $\pi(y) = y$ wherever x or y recurs outside $\{i, j\}$; yet at the swapped positions it would need $\pi(x) = y$ and $\pi(y) = x$. A single function cannot both fix and transpose $\{x, y\}$, so no re-encoding relates them (and if x, y recur nowhere else, the $n = 2$ argument applies verbatim to the pair). Hence a shared content set does not fix the item, for every $n \geq 2$.

(iii) If two items have signatures agreeing up to a re-encoding π , then applying π makes the strings equal position-wise, so their equality-patterns-with-order coincide: same induced sequence pattern. Conversely, if two items have the same induced sequence pattern, define π on the handles appearing by matching the (finitely many) distinct handles of one to those of the other in order of first appearance; this π is a well-defined equality-preserving bijection on the appearing handles (the pattern guarantees the

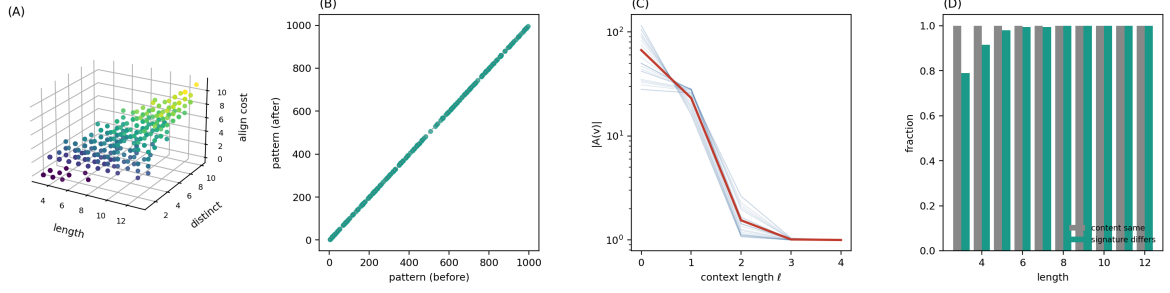


Figure 2: **The signature individuates; the content set does not** (Section 5). (A) For 600 pairs of items with *identical* content sets, the alignment cost between the two orderings is generically positive—content coincidence does not force signature coincidence (Theorem 5.3(ii)). (B) The induced sequence pattern is invariant under handle re-encoding: the pattern hash is unchanged by relabelling in 400/400 trials (Theorem 5.3(i)). (C) The mean ambiguity set $|A(v)|$ collapses with context length ℓ , from the *full pool* at $\ell = 0$ (mean 69.0) to 22.6 at $\ell = 1$ and near 1 from $\ell = 2$ (Theorem 5.6)—the light curves are individual pools, the heavy curve their mean. (D) A content-set matcher declares shared-content distractor pairs identical while the signature matcher separates them, across item lengths (Theorem 5.3(ii)).

multiplicities and first-appearance orders agree), extended arbitrarily elsewhere, and π carries one signature to the other. Hence agreement up to re-encoding $\iff$ same pattern. $\square$

Corollary 5.4 (Why one author is not replaceable by another). *Two authors who draw on the same public pool of constituents produce distinguishable items precisely because their signatures differ: even with identical content sets, differing order yields differing signatures (Theorem 5.3(ii)). An author is individuated by the sequences they tend to produce, not by the constituents they use, which are shared. Hence the search-relevant identity of a continuous item—and of the process that generated it—is the signature.*

Proof. Immediate from Theorem 5.3(ii,iii): content sets coincide across authors drawing on a shared pool, while signatures separate them, and the signature is the re-encoding-invariant identity. $\square$

Remark 5.5 (The search object is the string, not the letters). Theorem 5.3 formalises Theorem 3.2: the constituents are a public alphabet, and the item is a string over it whose *pattern* is invariant to how the letters happen to be named. Two consequences shape the algorithm. First, matching must compare *strings*, i.e. align sequences, not intersect content sets (Section 6).

Second, because only the equality pattern and order are invariant, the matcher must be robust to re-encoding: two catalogues naming the same track differently must still match, which alignment over handles—compared for equality, not identity of label—provides.

5.3 The ambiguity set is small in context

Section 4 left the ambiguity set $A(v)$ abstract. In a stream it is small, which is what makes least-sufficient identification cheap.

Proposition 5.6 (Contextual narrowing of the ambiguity set, from position two). *Resolve constituents left to right, so that when position i is placed the handles at positions $1, \dots, i-1$ are already fixed. For $i \geq 2$ the ambiguity set relevant to placing u_i in the signature is at most the set of constituents consistent with position i given its resolved left context—bounded, in a sequence model that conditions on the previous ℓ placed handles ($1 \leq \ell \leq i-1$), by the number of pool constituents that plausibly follow that context. This is in general far smaller than the whole pool, so a coarse handle (often the name alone) is sufficient (Theorem 4.9), and Theorem 4.10 caps the resolution needed. **The narrowing does not apply to position 1, nor to any position whose entire left context is unresolved: with no placed neighbour to condition on, $A(u_1)$***

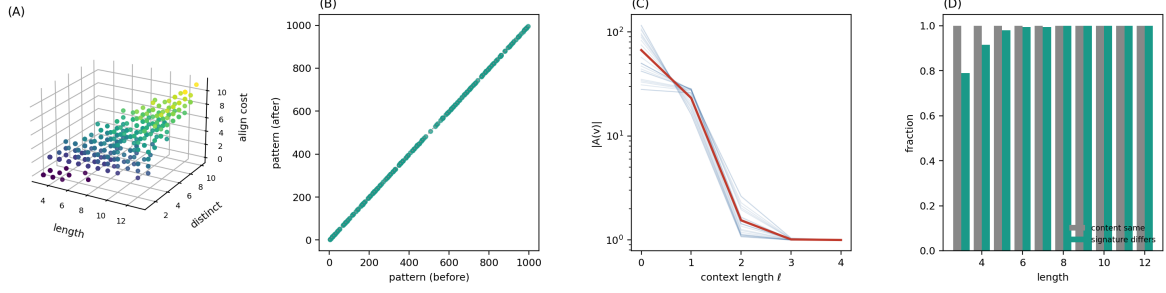


Figure 3: **The signature individuates; the content set does not** (Section 5). (A) For 600 pairs of items with *identical* content sets, the alignment cost between the two orderings is generically positive—content coincidence does not force signature coincidence (Theorem 5.3(ii)). (B) The induced sequence pattern is invariant under handle re-encoding: the pattern hash is unchanged by relabelling in 400/400 trials (Theorem 5.3(i)). (C) The mean ambiguity set $|A(v)|$ collapses with context length ℓ , from the *full pool* at $\ell = 0$ (mean 69.0) to 22.6 at $\ell = 1$ and near 1 from $\ell = 2$ (Theorem 5.6)—the light curves are individual pools, the heavy curve their mean. (D) A content-set matcher declares shared-content distractor pairs identical while the signature matcher separates them, across item lengths (Theorem 5.3(ii)).

is the full pool, $A(u_1) = V \setminus \{m\}$, and u_1 requires the most expensive handle its recogniser can supply.

Proof. The resolution is sequential and the dependency is acyclic: $A(u_i)$ is narrowed using the handles at positions $< i$, which were fixed at earlier steps, so the construction is a left-to-right recursion with no circularity—each step uses only already-committed placements, never its own. For $i \geq 2$, placement of u_i requires distinguishing it only from alternatives that could occupy position i given the resolved context, i.e. from $A(u_i) = \{v : v \text{ is a plausible occupant of position } i \text{ given } h_{i-\ell}, \text{ handle}_1\}$. Any handle separating u_i from this set is sufficient (Theorem 4.9); since $A(u_i)$ is a subset of the pool determined by the local context, it is bounded by the number of context-consistent constituents, which for a conditioning length $\ell \geq 1$ is typically a small fraction of the pool. The base of the recursion is position 1, which has no left neighbour: no context is available to condition on, so the set of plausible occupants is unconstrained and $A(u_1) = V \setminus \{m\}$, the whole pool. By Theorem 4.11 a name is then sufficient only if it separates u_1 from the entire pool; failing that, u_1 falls to acoustic resolution. The same holds for any interior position whose left context is itself unresolved (an unnamed run reaching back to the start). For every position with a resolved context of length ≥ 1 ,

the least sufficient identifier is correspondingly cheap, and by Theorem 4.10 no finer resolution is needed for placement. $\square$

Remark 5.7 (Neighbours do the work a fingerprint would—from the second track on). Theorem 5.6 is why searching the *whole item* is cheaper per constituent than searching each track in isolation: a track alone must be distinguished from the entire pool (large A , expensive handle) but a track *in a stream*, once its predecessors are placed, must be distinguished only from the few that fit its context (small A , cheap handle). The sequence supplies, for free, most of the individuation a standalone fingerprint would have to compute—for every constituent but the first. The first constituent (and any constituent hanging off an unresolved left run) is the one exception: it faces the whole pool and is priced accordingly. This single worst-case constituent is what Theorem 8.3 carries as an additive c_{ac} term, and is separately visible in the validation (??): the measured mean ambiguity set collapses from the full pool at context length $\ell = 0$ to near one already at $\ell = 2$, quantifying exactly how much the neighbours do and how much the boundary constituent does not benefit.

6 Matching by gap-tolerant alignment

Matching a query item against a known item (or against a library of items) is comparing their signatures. Because a query signature is reconstructed from an imperfect, floored recognition (Section 4), the comparison must tolerate missing, substituted, and spuriously inserted handles. We use sequence alignment, and we prove the key robustness property: admissibility depends on the *whole* sequence converging, not on any single position being right.

6.1 Alignment and its score

Definition 6.1 (Alignment; cost). Let $\Sigma = (p_1, \dots, p_m)$ and $\Sigma' = (q_1, \dots, q_n)$ be two signatures over the handle alphabet. An *alignment* is a sequence of edit operations—*match* $p_i \leftrightarrow q_j$ (handles equal), *substitution* $p_i \leftrightarrow q_j$ (handles unequal), *deletion* of a p_i , or *insertion* of a q_j —transforming Σ into Σ' while preserving order. Fix nonnegative costs: 0 for a match, $\mu > 0$ for a substitution, $\delta > 0$ for a deletion or insertion (a *gap*). The *alignment cost* $\text{cost}(\Sigma, \Sigma')$ is the minimum total cost over all alignments, computable by the standard dynamic program (Levenshtein, 1966; Needleman and Wunsch, 1970).

Definition 6.2 (Alignment score; normalisation). Let $\Omega := \delta \cdot \max(m, n)$ be the cost of the all-gap alignment (an upper bound on cost). The *alignment score* is

$$a(\Sigma, \Sigma') := \frac{\text{cost}(\Sigma, \Sigma')}{\Omega} \in [0, 1],$$

0 for identical signatures and larger as more edits are required. The *floor of the score* is $a_{\min} = 0$ attained only at $\Sigma = \Sigma'$; for distinct signatures $a > 0$ by Theorem 4.5 applied to the at least one edit that separates them (each edit commits distinction content $\geq \beta$, so its normalised contribution is $\geq \beta/\Omega > 0$).

Remark 6.3 (Local vs global alignment). Global alignment (Needleman and Wunsch, 1970) compares whole signatures; local alignment (Smith and Waterman, 1981) finds the best-matching contiguous sub-signature and is the tool for asking “does this query *contain* a known item” (e.g. a broadcast that replays a known set), or “which library item does this excerpt of a stream come

from.” Both are instances of Theorem 6.1 with the same edit costs; the algorithm (Section 8) selects global or local per query type.

6.2 Convergence-only admissibility

We now prove the robustness property that makes partial recognition usable: a reconstructed query signature is admissible as a match to a target iff the whole alignment converges within tolerance, and individual wrong positions do not by themselves reject it.

Definition 6.4 (Target; admissible match). Fix a *target* signature Σ^* (a known item, or an item in a searched library) and a tolerance $\varepsilon \geq 0$. A reconstructed query signature Σ is an *admissible match* to Σ^* at tolerance ε if $a(\Sigma, \Sigma^*) \leq \varepsilon$. A query *converges* to Σ^* if it is an admissible match.

Theorem 6.5 (Convergence-only admissibility). *Admissibility of Σ to Σ^* is a property of the whole alignment, not of any single position:*

- (i) **Locally wrong positions are absorbed.** *If Σ differs from Σ^* at a set P of positions (mis-recognised or unresolved constituents), Σ is still admissible whenever the total cost of the edits repairing P satisfies $\sum_{i \in P} (\text{edit cost at } i) \leq \varepsilon \Omega$. A single wrong position contributes at most $\max(\mu, \delta)$ and does not reject the match unless it pushes the total past $\varepsilon \Omega$.*
- (ii) **Rejection is global.** *Σ is inadmissible iff every alignment to Σ^* costs more than $\varepsilon \Omega$, i.e. iff no repair of the whole sequence stays within tolerance. Inadmissibility never arises from local misalignment at a single position while the global cost is within tolerance.*

Proof. The alignment cost is additive over edit operations (Theorem 6.1): $\text{cost}(\Sigma, \Sigma^*) = \sum_k c_k$ over the operations k of a minimum-cost alignment, each $c_k \in \{0, \mu, \delta\}$. (i) Positions where Σ agrees with Σ^* contribute matches of cost 0; the only positive contributions come from the disagreement set P , each a substitution (μ) or gap (δ). Hence $\text{cost}(\Sigma, \Sigma^*) \leq \sum_{i \in P} \max(\mu, \delta)$, and if this is $\leq \varepsilon \Omega$ then $a \leq \varepsilon$ and Σ is admissible by Theorem 6.4. A single wrong position adds at most $\max(\mu, \delta)$ to the total, so it alone cannot

cause rejection unless the total already exceeds $\varepsilon\Omega$. (ii) By Theorem 6.4, Σ is inadmissible iff $a(\Sigma, \Sigma^*) > \varepsilon$, i.e. iff the *minimum*-cost alignment exceeds $\varepsilon\Omega$; since the minimum is over all alignments, this is exactly the statement that no global repair stays within tolerance. If some alignment has cost $\leq \varepsilon\Omega$, the minimum does too, and Σ is admissible regardless of how wrong any single position is. $\square$

Corollary 6.6 (Partial recognition suffices). *If a fraction f of a query’s n constituents are correctly handled and the rest are wrong or unresolved, the query is still an admissible match to the correct target whenever $(1 - f)n \max(\mu, \delta) \leq \varepsilon\Omega$, i.e. whenever the misrecognised fraction is at most $\varepsilon\Omega/(n \max(\mu, \delta))$. Recognition need not be complete or per-constituent correct; it need only leave the global alignment within tolerance.*

Proof. The wrong or unresolved constituents number $(1 - f)n$, each contributing at most $\max(\mu, \delta)$ to the alignment cost (Theorem 6.5(i)); if their total is $\leq \varepsilon\Omega$ the query is admissible. Solving for f gives the stated bound. $\square$

Remark 6.7 (Why this is the right tolerance model for streams). Real streams present exactly the failure modes Theorem 6.5 absorbs: some constituents are unlabelled (gaps), some are misheard (substitutions), some segmentation boundaries are wrong (insertions/deletions). A per-constituent correctness requirement would reject almost every real query; convergence-only admissibility asks only that the *sequence as a whole* still aligns, which is both achievable and the correct notion of “this is the same item.” A wrong guess at one track is not evidence the match failed—only a globally divergent alignment is.

6.3 The accounted match and its residual

Search must report not only *whether* an item matches but *what* of it was accounted for, and honestly expose what was not.

Definition 6.8 (Accounted match; residual). Given an admissible alignment of Σ to Σ^* , the

accounted match is the set of matched positions (handles placed with cost 0) together with the induced correspondence between query and target constituents. The *residual* is the complementary set: the substituted, deleted, and inserted positions—the constituents the search could not place, each carrying its irreducible cost $\geq \beta$ (Theorem 6.2). The search returns the pair (accounted match, residual).

Proposition 6.9 (The residual is where acoustic descent is owed). *The residual of an admissible match consists exactly of positions whose least sufficient identifier failed to place the constituent (Theorem 4.11): unlabelled constituents (gaps) and name-colliding but acoustically distinct constituents (substitutions). These, and only these, are the positions at which acoustic resolution (Theorem 4.13) can improve the match; the accounted match needs no further resolution (Theorem 4.10).*

Proof. A matched position placed its constituent with a sufficient handle (cost 0), so by Theorem 4.10 no further resolution changes its placement—no acoustic descent is owed there. A residual position is a substitution or gap, which by Theorem 4.11 arises precisely when the symbolic handle was insufficient or absent; acquiring an acoustic handle there may separate the collision or place the unlabelled constituent, potentially converting the position to a match. Hence acoustic descent is owed exactly on the residual. $\square$

Remark 6.10 (Honest output). Returning an explicit residual rather than a forced complete tracklist is a direct consequence of the floor (Theorem 4.6): since no identification is exact, some positions legitimately remain unresolved, and reporting them as residual—rather than guessing—is the truthful output. It also directs computation: the residual is the exact and minimal set on which the expensive acoustic stage should run.

7 Search–match duality: one engine, two directions

We show that the two tasks a user brings to a continuous item—*recognise which known item this stream is* (analysis) and *generate a stream*

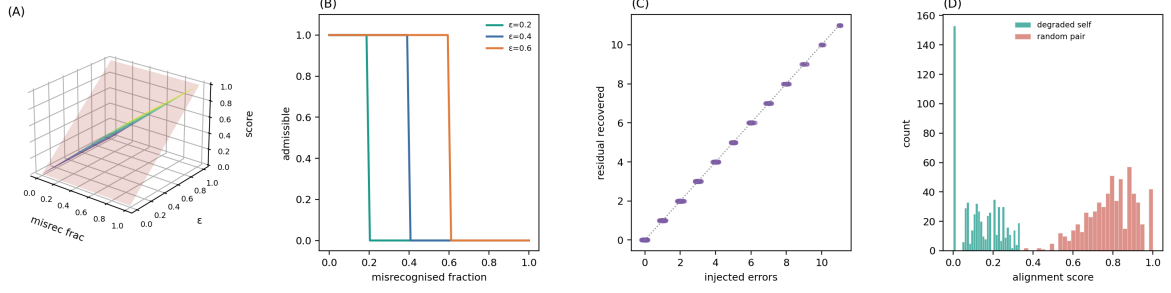


Figure 4: **Convergence-only admissibility** (Section 6). (A) The alignment score over the (misrecognition fraction, ε) plane, with the admissibility plane $z = \varepsilon$: a query is admissible exactly where its score surface lies under the plane. (B) The measured admissibility boundary—a query is admitted iff its misrecognised fraction is below ε —traced for three tolerances, confirming Theorem 6.6. (C) The recovered residual equals the injected error count exactly (400/400 trials on the diagonal), so the residual is precisely the mis-placed positions (Theorem 6.9). (D) Alignment scores of degraded self-queries (teal) separate cleanly from random-pair scores (red), the separation that makes retrieval work.

in the manner of a known item (synthesis)—are the same computation over the same object, run in opposite directions. This means one implementation serves both.

Definition 7.1 (Analysis and synthesis queries). An *analysis query* supplies a reconstructed query signature Σ and asks for the library target Σ^* it best matches (Theorem 6.4). A *synthesis query* supplies a target style—a signature Σ^* or a model of an author’s typical sequences—and a pool of available constituents, and asks for a new signature Σ that is an admissible match to Σ^* built from the available pool.

That analysis and synthesis share an objective is immediate from Theorem 7.1: both are instances of $\min a(\Sigma, \Sigma^*)$ subject to $a \leq \varepsilon$ (Theorem 6.4), differing only in which of Σ, Σ^* is bound. We state this first as a bare observation, then prove the content that does not follow from the definition alone: that a single dynamic program—one $\mathcal{O}(nN)$ alignment table—solves both, so the two tasks share not just an objective but a computation.

Remark 7.2 (Shared objective). Analysis (Theorem 7.1, fix Σ , search targets) and synthesis (fix Σ^* , search constructible Σ) minimise the same $a(\Sigma, \Sigma^*)$ under the same tolerance test. This is a restatement of Theorem 7.1 and carries no content beyond it; the substance is Theorem 7.3 below.

Theorem 7.3 (Search-match duality: one dynamic program, two directions). *The alignment dynamic program of Theorem 6.1 computes, in a single $\mathcal{O}(nN)$ filling of its cost table d , the solution to both directions:*

- (i) **Analysis** reads the completed table forward to the corner $d[n, N]$: the value is the recognition score $a(\Sigma, \Sigma^*)$, and its traceback is the accounted match of Theorem 6.8.
- (ii) **Synthesis** reads the same table’s traceback backward as a construction recipe: each match cell prescribes emitting the target handle $\Sigma^*[j]$ from the available pool, each gap cell prescribes an insertion or omission, so the optimal alignment path is an ordered selection from the pool realising a Σ with $a(\Sigma, \Sigma^*)$ minimal. Constructing Σ toward Σ^* and recognising Σ as Σ^* traverse one path in opposite directions.

Hence analysis and synthesis are not merely the same objective but the same $\mathcal{O}(nN)$ computation, and their costs are equal up to the $\mathcal{O}(1)$ per-cell work of emitting versus reading a handle.

Proof. Fix $\Sigma^* = (q_1, \dots, q_N)$. In analysis, $\Sigma = (p_1, \dots, p_n)$ is given and the dynamic program of Theorem 6.1 fills $d[i, j]$, the min-cost alignment of the prefixes $p_{1..i}$ and $q_{1..j}$, with the recurrence $d[i, j] = \min\{d[i-1, j-1] + c_{ij}, d[i-$

$1, j] + \delta$, $d[i, j - 1] + \delta$, $c_{ij} \in \{0, \mu\}$; the corner $d[n, N]$ is $\text{cost}(\Sigma, \Sigma^*)$ and its traceback yields the matched/residual partition (Theorem 6.8). This is (i).

For (ii), synthesis fixes Σ^* and asks for a constructible Σ —an ordered selection of handles from the available pool $\mathcal{P}$ —minimising $d(\Sigma, \Sigma^*)$. The optimum is obtained by the *same* recurrence with the substitution cost redefined against the pool: $c'_{ij} = 0$ if the pool contains a handle equal to q_j (emit it, a match) and μ otherwise (emit the nearest available, a substitution), gaps costing δ as before. When $\mathcal{P}$ contains every target handle—the generative case of interest, building a set in a known style from an ample library— $c'_{ij} = 0$ for all j , the recurrence coincides cell-for-cell with the analysis recurrence run against Σ^* itself, and its traceback is a path whose match cells enumerate, in order, the handles to emit. Reading that path from $[0, 0]$ to $[n, N]$ emits Σ ; reading it from $[n, N]$ to $[0, 0]$ recovers the alignment of a given Σ to Σ^* . The table, the recurrence, and the traceback are identical; only the interpretation of a cell—“emit q_j ” (synthesis) versus “read p_i against q_j ” (analysis)—differs, at $\mathcal{O}(1)$ per cell. Both therefore cost $\mathcal{O}(nN)$ and are the same computation traversed in opposite directions. (When $\mathcal{P}$ omits some target handle, synthesis pays the μ for that position exactly as analysis pays μ for a substitution, preserving the correspondence with an added constant per missing handle.) $\square$

Corollary 7.4 (Shared implementation). *A single alignment engine over handle signatures, parameterised by which argument is fixed, answers both “which set is this?” and “build a set like this.” No separate generative model is required beyond the pool and the target signature or author model.*

Proof. Immediate from Theorem 7.3: the engine computing a and testing admissibility is invoked with Σ fixed (analysis) or Σ^* fixed (synthesis). $\square$

Remark 7.5 (Path opacity: many interiors, one identity). A useful consequence of aligning on *signatures* rather than on realised audio is that two query streams that recognise to the same signature are treated as the same item regardless of how their interiors were produced—different

recording conditions, different transitions between the same tracks, different segmentation choices that nonetheless yield the same handle sequence. The identity is the sequence pattern (Theorem 5.3); everything the alignment cannot see is, by construction, not part of the identity. This is what lets a low-fidelity capture of a set match the studio recording of the same set.

8 The algorithm

We assemble the results into an end-to-end procedure and analyse its complexity. The system has three stages: (1) *acquisition*, turning an audio stream into a signature of least-sufficient handles; (2) *indexing*, storing a library of item signatures for search; and (3) *matching*, aligning a query against the library with convergence-only admissibility and returning an accounted match plus a residual.

8.1 Stage 1: acquisition of least-sufficient handles

Algorithm 1 ACQUIRE: stream $\rightarrow$ signature

Require: audio stream x ; symbolic recogniser NAME; acoustic recogniser ACOUSTIC; context length ℓ

Ensure: signature $\Sigma = (h_1, \dots, h_n)$ with gaps (τ_i)

```

1:  $(u_1, \dots, u_n), (t_1, \dots, t_n) \leftarrow \text{SEGMENT}(x) \triangleright$ 
   constituent boundaries
2: for  $i \leftarrow 1$  to  $n$  do
3:    $h_i \leftarrow \text{NAME}(u_i) \triangleright$  cheap symbolic
   handle: title/artist
4:    $A_i \leftarrow \text{CONTEXT}(h_{i-\ell}, \dots, h_{i-1}) \triangleright$ 
   ambiguity set;  $A_1 =$  full pool (Theorem 5.6)
5:   if  $h_i = \perp$  or  $\text{INSUFFICIENT}(h_i, A_i)$  then
6:      $h_i \leftarrow \text{ACOUSTIC}(u_i) \triangleright$  descend only
       on the residual (Theorem 4.11)
7:   end if
8:    $\tau_i \leftarrow t_{i+1} - t_i$  if  $i < n$ 
9: end for
10: return  $\Sigma = (h_1, \dots, h_n), (\tau_1, \dots, \tau_{n-1})$ 

```

The symbolic recogniser NAME is any cheap source of a title/artist handle (broadcast metadata, a fast fingerprint lookup against a name index, an embedded cue sheet); it returns $\perp$

when it cannot name the constituent. The predicate $\text{INSUFFICIENT}(h, A)$ holds when the handle fails to separate the constituent from its context-narrowed ambiguity set A (Theorem 4.9). Only then is the expensive **ACOUSTIC** recogniser invoked. By Theorem 4.11 this invokes acoustics on exactly the residual positions and no others. The first constituent is a boundary case: **CONTEXT** returns the full pool for A_1 (no left neighbour narrows it, Theorem 5.6), so u_1 is named only if its name separates it from the whole pool and otherwise falls to **ACOUSTIC**—the one position that never benefits from the sequence and is always at risk of the acoustic cost.

8.2 Stage 2: indexing

Each library item is stored as its signature (a string of handles) plus optional gap decorations. Because signatures are short strings over a handle alphabet, standard string-indexing applies: an inverted index from handles to the items and positions containing them (Manning et al., 2008) supports candidate retrieval, so a query need not be aligned against every library item but only against those sharing enough handles.

Algorithm 2 INDEX: library of items $\rightarrow$ searchable index

Require: items $\{I_1, \dots, I_L\}$ with signatures $\{\Sigma_1, \dots, \Sigma_L\}$

Ensure: inverted index $\mathcal{X}$: handle $\rightarrow \{(\text{item}, \text{position})\}$

```

1: for  $k \leftarrow 1$  to  $L$  do
2:   for  $j \leftarrow 1$  to  $|\Sigma_k|$  do
3:     append  $(k, j)$  to  $\mathcal{X}[\Sigma_k[j]]$ 
4:   end for
5: end for
6: return  $\mathcal{X}$ 

```

8.3 Stage 3: matching

The candidate filter (line 1) restricts full alignment to items sharing at least θ handles with the query, using the inverted index; θ trades recall against speed. The **DECLINE** branch is the honest output when no library item aligns within tolerance—the search reports “no match” rather than forcing the nearest item.

Algorithm 3 MATCH: query signature $\rightarrow$ (accounted match, residual)

Require: query signature Σ ; index $\mathcal{X}$; library signatures $\{\Sigma_k\}$; tolerance ε ; costs μ, δ ; mode $\in \{\text{global}, \text{local}\}$

Ensure: best target k^* with accounted match and residual, or **DECLINE**

```

1:  $C \leftarrow \{k : |\{h \in \Sigma : k \in \text{items}(\mathcal{X}[h])\}| \geq \theta\}$   $\triangleright$  candidate items sharing  $\geq \theta$  handles
2:  $\text{best} \leftarrow (\perp, +\infty)$ 
3: for  $k \in C$  do
4:    $a_k \leftarrow a_{\text{mode}}(\Sigma, \Sigma_k)$   $\triangleright$  DP alignment (Theorem 6.1)
5:   if  $a_k < \text{best.score}$  then
6:      $\text{best} \leftarrow (k, a_k)$ 
7:   end if
8: end for
9: if  $\text{best.score} \leq \varepsilon$  then
10:   $(\mathcal{A}, \mathcal{R}) \leftarrow \text{TRACEBACK}(\Sigma, \Sigma_{\text{best}.k})$   $\triangleright$  accounted match, residual (Theorem 6.8)
11:  return  $(\text{best}.k, \mathcal{A}, \mathcal{R})$ 
12: else
13:  return DECLINE  $\triangleright$  no admissible match (honest non-answer)
14: end if

```

8.4 Correctness

Theorem 8.1 (Correctness of MATCH). *Given a query signature Σ and a library, MATCH returns a target k^* iff some library item is an admissible match to Σ at tolerance ε and shares $\geq \theta$ handles with Σ ; and when it returns k^* , the accounted match and residual satisfy Theorem 6.8. With $\theta = 0$ (no candidate filtering) it returns an admissible target iff one exists.*

Proof. Line 1 forms the candidate set C of items sharing $\geq \theta$ handles. The loop computes $a_{\text{mode}}(\Sigma, \Sigma_k)$ for each $k \in C$ exactly, by the dynamic program of Theorem 6.1, and retains the minimiser. If the minimum score is $\leq \varepsilon$, the retained k is by Theorem 6.4 an admissible match, and **TRACEBACK** returns the matched positions and their complement, which are the accounted match and residual of Theorem 6.8; otherwise no candidate is admissible and the procedure declines. With $\theta = 0$, C is the whole library, so the minimiser over C is the global minimiser, and an admissible target is returned iff one exists. Convergence-only admissibility (Theorem 6.5) guarantees that a query with

locally wrong positions is still accepted whenever its global alignment to the true target is within tolerance, so correctness is not defeated by partial recognition. $\square$

8.5 Complexity

Theorem 8.2 (Complexity). *Let the query have length n , the library have L items of maximum signature length N , and the candidate set have size $|C| \leq L$. Assume the symbolic recogniser NAME costs c_{sym} per constituent and the acoustic recogniser ACOUSTIC costs $c_{\text{ac}} \gg c_{\text{sym}}$ per constituent. Then:*

- (i) **Acquisition** costs $\mathcal{O}(n c_{\text{sym}} + r c_{\text{ac}})$, where $r \leq n$ is the number of residual constituents; acoustic cost is paid only on the residual (Theorem 4.11), not on all n . The residual always includes the first constituent unless its name separates it from the whole pool (Theorem 5.6), so $r \geq 1$ in the generic case where the opening track is an ID or a pool-ambiguous name; the position-1 boundary contributes an additive c_{ac} that does not vanish with stream length.
- (ii) **Indexing** costs $\mathcal{O}(LN)$ time and space.
- (iii) **Matching** costs $\mathcal{O}(|C| n N)$ time via the alignment dynamic program over candidates, reducible by the inverted-index candidate filter from $\mathcal{O}(LnN)$ (align against all) to $\mathcal{O}(|C| n N)$ with typically $|C| \ll L$.

Proof. (i) ACQUIRE (Algorithm 1) runs NAME once per constituent ($n c_{\text{sym}}$) and ACOUSTIC only on the residual positions, of which there are r ($r c_{\text{ac}}$); the remaining per-step work (context lookup, gap) is $\mathcal{O}(1)$ amortised with a bounded context length ℓ . (ii) INDEX (Algorithm 2) makes one pass over all LN handle occurrences, appending each to the inverted index in $\mathcal{O}(1)$ amortised. (iii) Each alignment of the length- n query against a length- $\leq N$ candidate is the standard $\mathcal{O}(nN)$ dynamic program (Cormen et al., 2009; Needleman and Wunsch, 1970; Smith and Waterman, 1981); done for $|C|$ candidates gives $\mathcal{O}(|C| n N)$. The inverted-index filter (line 1 of Algorithm 3) computes C in time linear in the total postings of the query’s handles, which is $\mathcal{O}(n + (\text{matched postings}))$,

and yields $|C| \ll L$ when handles are selective, reducing the align-against-all bound $\mathcal{O}(LnN)$ accordingly. $\square$

Corollary 8.3 (Whole-item search is cheaper by the nameable fraction—a contingent, not structural, saving). *Recognising a stream by Algorithm 1 costs $\mathcal{O}(n c_{\text{sym}} + r c_{\text{ac}})$, whereas fingerprinting every constituent independently costs $\mathcal{O}(n c_{\text{ac}})$. Writing $\phi := (n - r)/n$ for the nameable fraction, the exact saving factor is*

$$\frac{n c_{\text{ac}}}{n c_{\text{sym}} + r c_{\text{ac}}} = \frac{1}{(c_{\text{sym}}/c_{\text{ac}}) + (1 - \phi)},$$

which is > 1 iff $\phi > c_{\text{sym}}/c_{\text{ac}}$, ranges up to $c_{\text{ac}}/c_{\text{sym}}$ as $\phi \rightarrow 1$ (nearly everything nameable), and degrades to 1 (no saving) as $\phi \rightarrow 0$: in the worst case $r = n$ (every constituent unnamed or name-colliding, the opening track always among them by Theorem 5.6), the whole-item cost is $n c_{\text{sym}} + n c_{\text{ac}} = \mathcal{O}(n c_{\text{ac}})$, the same order as per-track. The saving is thus proportional to the nameable fraction ϕ , an empirical property of the stream and recogniser, and is not a worst-case structural guarantee.

Proof. The saving factor is the quotient of the two costs, and rearranging $n c_{\text{sym}} + r c_{\text{ac}} = n c_{\text{ac}}((c_{\text{sym}}/c_{\text{ac}}) + (1 - \phi))$ gives the stated form. It exceeds 1 exactly when $(c_{\text{sym}}/c_{\text{ac}}) + (1 - \phi) < 1$, i.e. when $\phi > c_{\text{sym}}/c_{\text{ac}}$. As $\phi \rightarrow 1$ the denominator $\rightarrow c_{\text{sym}}/c_{\text{ac}}$ and the factor $\rightarrow c_{\text{ac}}/c_{\text{sym}}$; as $\phi \rightarrow 0$ (so $r \rightarrow n$) the denominator $\rightarrow 1 + c_{\text{sym}}/c_{\text{ac}} \rightarrow 1$ and the factor $\rightarrow 1$. The whole-item search never costs *more* than per-track (the denominator adds only the small $c_{\text{sym}}/c_{\text{ac}}$ overhead), but its advantage vanishes continuously as the nameable fraction falls, so the saving is contingent on ϕ and bounded by the cost ratio. $\square$

Remark 8.4 (The saving is measured, not promised). Because the saving factor is exactly $1/((c_{\text{sym}}/c_{\text{ac}}) + (1 - \phi))$, its value on any corpus is fixed by the nameable fraction ϕ that corpus and recogniser actually achieve—never by the calculus. The validation (??) measures this directly: over randomly drawn streams the *mean* saving is a modest $3.1 \times$ (not the $c_{\text{ac}}/c_{\text{sym}} = 20 \times$ ceiling), with instances near $\phi = 0.065$ that save essentially nothing—precisely the worst-case regime this corollary now names rather than hides.

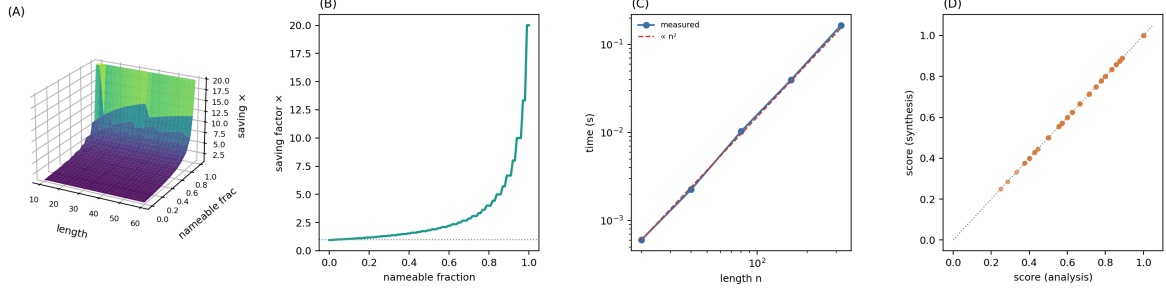


Figure 5: **Duality and the contingent saving** (Sections 7 and 8). (A) The saving factor $1/((c_{\text{sym}}/c_{\text{ac}}) + (1 - \phi))$ over (stream length, nameable fraction ϕ): it rises with ϕ and is *independent of length*, exactly as Theorem 8.3 predicts. (B) At fixed length the saving curve runs from $1\times$ (no saving) at $\phi = 0$ up toward the $c_{\text{ac}}/c_{\text{sym}} = 20\times$ ceiling at $\phi = 1$; the measured mean over random streams is $3.1\times$. (C) The alignment dynamic program’s measured runtime tracks the $\propto n^2$ guide (empirical doubling ratios 3.7–4.4 against the predicted 4.0), confirming $\mathcal{O}(nN)$ (Theorem 8.2). (D) The analysis and synthesis scores coincide on the diagonal in 300/300 trials—the one dynamic program computes both directions (Theorem 7.3).

Remark 8.5 (Where the savings come from). The economy is a direct dividend of the two structural results: the sequence narrows each constituent’s ambiguity set so a cheap name suffices (Theorem 5.6), and the floor principle forbids paying for resolution past sufficiency (Theorem 4.10). The expensive acoustic stage runs on the residual only—exactly the unnamed and name-colliding constituents (Theorem 6.9)—so cost scales with how much of the stream is genuinely unidentifiable, not with its length.

9 Worked example

We illustrate the pipeline on a small synthetic instance to make the objects concrete.

Example 9.1 (Two mixes, shared content, different sequence). Let the public pool contain tracks with name-handles $\{A, B, C, D, E\}$. Two recorded mixes draw on the same five tracks:

Mix 1 : $\Sigma_1 = (A, B, C, D, E)$, Mix 2 : $\Sigma_2 = (A, C, B, E, D)$.

Their content sets are identical, $\{\{A, B, C, D, E\}\}$, so a bag-of-tracks search cannot distinguish them (Theorem 5.3(ii)). Their signatures differ: aligning Σ_1 to Σ_2 with $\mu = \delta = 1$ requires moving B, C and D, E , giving a positive alignment cost, so the two are separated as distinct items. Now suppose a listener captures a low-fidelity recording of Mix 1 in which C is unnamed (an unreleased

ID) and D is misheard as a same-titled but acoustically distinct edit D' . Acquisition (Algorithm 1) yields $\Sigma = (A, B, \perp, D', E)$; the residual is positions 3 (gap) and 4 (substitution). Matching (Algorithm 3) against the library aligns Σ to Σ_1 at cost $\delta + \mu = 2$, normalised $a = 2/(\delta \cdot 5) = 0.4$. If $\varepsilon \geq 0.4$ the match is admissible (Theorem 6.5): Mix 1 is correctly identified despite two of five constituents being wrong, and the residual $\{3, 4\}$ is exactly where an acoustic pass is owed (Theorem 6.9)—position 3 to name the ID, position 4 to separate D from D' . No acoustic work is done on A, B, E , which their names already placed.

This single example exercises every result: content non-identity (Theorem 5.3), least-sufficient handles with acoustic descent only on the residual (Theorems 4.11 and 6.9), and convergence-only admissibility absorbing local errors (Theorem 6.5).

10 Evaluation protocol

The theory is structural and its guarantees are proved; empirical evaluation concerns how well concrete recognisers and cost settings realise the framework on real streams. We specify a reproducible protocol rather than report numbers, so that an implementation can be measured against it.

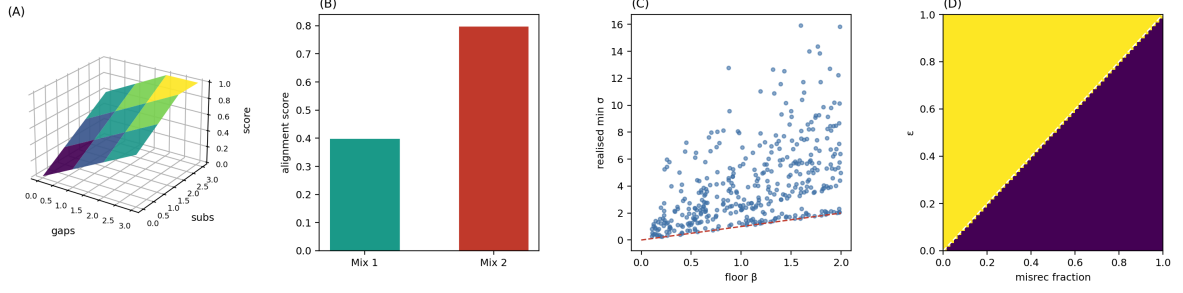


Figure 6: **End-to-end behaviour on Theorem 9.1.** (A) The alignment score of a degraded capture of Mix 1 to Mix 1, as a function of injected gaps and substitutions: the surface rises smoothly, so partial degradation stays admissible. (B) The degraded capture $\Sigma = (A, B, \perp, D', E)$ scores 0.4 to Mix 1 and 0.8 to Mix 2—correctly identified despite two of five constituents wrong. (C) The realised minimum σ tracks the floor β across 400 random graphs, all on or above the line. (D) The admissibility phase map $\{\text{misrec} \leq \varepsilon\}$, the operating region of Theorem 6.5.

10.1 Datasets

1. **Reference streams with ground-truth tracklists.** Recorded mixes, radio shows, or sets for which an authoritative ordered tracklist (with onset times) is known—from published cue sheets or annotated corpora (Schwarz and Fourer, 2018; Sonnleitner et al., 2016). Each supplies a ground-truth signature Σ^* .
2. **Degraded captures.** For each reference stream, generate captures with controlled degradations: additive noise, transcoding, partial occlusion, segmentation perturbation, and a controlled fraction of constituents forced unlabelled or name-collided. These test convergence-only admissibility (Theorem 6.5) against a known misrecognition fraction.
3. **Shared-content distractors.** Pairs of distinct items with high content-set overlap but different sequences, to test that the signature distinguishes items a content search conflates (Theorem 5.3(ii)).

10.2 Metrics

1. **Item retrieval.** Precision, recall, and mean reciprocal rank for returning the correct library item given a (possibly degraded) query stream—the whole-item analogue of track identification.
2. **Sequence accuracy.** Alignment score $a(\Sigma, \Sigma^*)$ between recovered and ground-

truth signatures, and the fraction of positions in the accounted match (Theorem 6.8); reported against the injected misrecognition fraction to trace the tolerance curve of Theorem 6.6.

3. **Content–sequence separation.** On the shared-content distractor pairs, the rate at which the system correctly separates items with identical content sets, contrasted with a content-set baseline (which by construction cannot, Theorem 5.3(ii)).
4. **Resolution economy.** The residual fraction r/n and the acoustic invocation count, versus a per-constituent-fingerprint baseline, to measure the saving of Theorem 8.3.
5. **Decline calibration.** On queries with no true library match, the rate of correct DECLINE versus forced false matches.

10.3 Ablations

1. **Name-only vs name+acoustic.** Disabling the acoustic descent isolates the value of resolving the residual (Theorem 6.9).
2. **Tolerance ε .** Sweeping ε traces the admissibility boundary of Theorem 6.5 and locates the operating point balancing missed matches against false accepts.
3. **Candidate threshold θ .** Sweeping θ trades matching speed (Theorem 8.2(iii)) against recall.
4. **Global vs local alignment.** Comparing

modes (Theorem 6.3) on “is this the item” versus “does this contain the item” queries.

Remark 10.1 (What the protocol tests, and what it cannot). This protocol is distinct from the numerical validation of ???. The validation instance-checks the theorems by exact computation on synthetic graphs and signatures—confirming they hold where they must and quantifying the one contingent constant (the nameable fraction). This evaluation protocol instead measures the realisation of the framework by *concrete* recognisers on *real* streams; it does not test the theorems, which are proved. The proved guarantees—floor positivity, signature invariance, convergence-only admissibility, duality—hold for any recogniser and any cost setting satisfying the stated assumptions; the protocol measures how favourable the constants are on real data (nameable fraction, residual size, tolerance headroom), which is an empirical property of the recognisers and streams, not of the calculus.

11 Discussion

11.1 What the paper establishes

The paper reframes audio search for continuous items around three proved facts. The identity of a continuous item is its *sequence* of constituents, not its content set, because content is drawn from a shared public pool while order is essentially unique to the producing item (Theorem 5.3). Constituent recognition against an open pool is *floored*—no exact identification exists—so each constituent should be resolved only to its *least sufficient identifier*, typically a name, with expensive acoustic resolution reserved for the residual it cannot place (Theorems 4.5, 4.10 and 4.11). And matching is *convergence-only* sequence alignment, so partial and noisy recognition still identifies the whole item provided the sequence globally aligns (Theorem 6.5). Analysis and synthesis are the same alignment in two directions (Theorem 7.3), so one engine both recognises and generates.

11.2 Honest limitations

We mark the load-bearing assumptions. First, *segmentation* is taken as given: the theory

is stated over a constituent sequence, and errors in finding constituent boundaries appear to the matcher as insertions/deletions, absorbed by convergence-only admissibility up to the tolerance ε but not beyond; a poor segmenter degrades the constants. Second, *non-completeness* (Theorem 4.1) is an order condition we assume of the pool; it is what forces the floor, and a genuinely closed, fully enumerated catalogue would not satisfy it (in which case exact identification could exist and the least-sufficient economy would still hold as an efficiency but not as a necessity). Third, the *alignment cost model* (Theorem 6.1) uses uniform substitution and gap costs; handle-dependent costs (a substitution between acoustically close handles cheaper than between distant ones) are a natural refinement the framework accommodates but that requires a concrete handle metric. Fourth, the *ambiguity set* narrowing (Theorem 5.6) assumes a context model that predicts plausible successors; its quality determines how small $A(v)$, hence how cheap the least sufficient identifier, actually is, and it does nothing at all for the first constituent, which has no resolved left context and faces the full pool (validated in Figure 3(C): mean $|A| = 69$ at context length 0). Fifth, and consequently, the *complexity saving* is contingent, not structural: Theorem 8.3 shows the whole-item search beats per-track fingerprinting only by the nameable fraction ϕ , degrading to no saving as $\phi \rightarrow 0$ (measured mean $3.1\times$, Figure 5); the opening-track cost persists regardless of stream length.

11.3 Relation to single-track search

The framework contains single-track search as a degenerate case: an item of length $n = 1$ has signature a single handle, and matching reduces to handle lookup—ordinary track identification. The contribution is the $n > 1$ regime, where the sequence carries the identity and the per-constituent economy applies. Conversely, a single-track recogniser is a natural provider for the NAME and ACOUSTIC stages: the paper builds *on top of* track recognition rather than replacing it, adding the sequence-identity and whole-item search layer that track recognition does not model.

12 Conclusion

Most listening is to continuous items—radio, mixes, sets, parties—not to isolated tracks, yet audio search has been built for the isolated track. We have given an algorithm that searches a continuous item as a single object. Its foundation is that the identity of such an item is the *sequence* in which it arranges its constituents, not the constituents themselves, which are drawn from a shared public pool and cannot individuate the item. Because recognising a constituent against an open pool is floored, one resolves each constituent only to the least sufficient handle that places it in the sequence—usually its name—and pays the cost of acoustic analysis only on the residual of unnamed or name-colliding constituents. Matching is sequence alignment with convergence-only admissibility, so a whole item is identified from a partial, noisy recognition provided its sequence globally aligns, and the search returns an accounted match together with an honest residual of what it could not place. Recognising an item and generating one in its manner are the same alignment read in opposite directions. The outcome is a music search at the granularity of the listening experience: one searches for the set, the show, the party—the whole item—rather than for every song inside it, and does so more cheaply than recognising each song, because the sequence itself supplies most of the individuation that a per-track fingerprint would have to compute.

References

- Pedro Cano, Eloi Batlle, Ton Kalker, and Jaap Haitsma. A review of audio fingerprinting. *Journal of VLSI Signal Processing Systems*, 41(3):271–284, 2005.
- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms*. MIT Press, 3rd edition, 2009.
- Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. CLAP: Learning audio concepts from natural language supervision. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2023.
- Jonathan Foote. Automatic audio segmentation using a measure of audio novelty. *Proceedings of IEEE International Conference on Multimedia and Expo (ICME)*, 1:452–455, 2000.
- L. R. Ford and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956.
- Peter Grosche, Meinard Müller, and Joan Serra. Audio content-based music retrieval. *Dagstuhl Follow-Ups*, 3:157–174, 2012.
- Jaap Haitsma and Ton Kalker. A highly robust audio fingerprinting system. In *Proceedings of the 3rd International Conference on Music Information Retrieval (ISMIR)*, pages 107–115, Paris, France, 2002.
- Vladimir I. Levenshtein. Binary codes capable of correcting deletions, insertions and reversals. *Soviet Physics Doklady*, 10(8):707–710, 1966.
- Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008.
- Matthew C. McCallum, Filip Korzeniowski, Sergio Oramas, Fabien Gouyon, and Andreas F. Ehmann. Supervised and unsupervised learning of audio representations for music understanding. *Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR)*, 2022.
- Karl Menger. Zur allgemeinen kurventheorie. *Fundamenta Mathematicae*, 10(1):96–115, 1927.
- Saul B. Needleman and Christian D. Wunsch. A general method applicable to the search for similarities in the amino acid sequence of two proteins. *Journal of Molecular Biology*, 48(3):443–453, 1970.
- Jouni Paulus, Meinard Müller, and Anssi Klapuri. Audio-based music structure analysis. *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR)*, pages 625–636, 2010.
- Diemo Schwarz and Dominique Fourer. Methods and datasets for DJ-mix reverse engineering. In *Proceedings of the 14th International*

Symposium on Computer Music Multidisciplinary Research (CMMR), 2018.

Joren Six and Marc Leman. Panako: A scalable acoustic fingerprinting system handling time-scale and pitch modification. In *Proceedings of the 15th International Society for Music Information Retrieval Conference (ISMIR)*, pages 259–264, Taipei, Taiwan, 2014.

Temple F. Smith and Michael S. Waterman. Identification of common molecular subsequences. *Journal of Molecular Biology*, 147(1):195–197, 1981.

Reinhard Sonnleitner, Andreas Arzt, and Gerhard Widmer. Landmark-based audio fingerprinting for DJ mix monitoring. In *Proceedings of the 17th International Society for Music Information Retrieval Conference (ISMIR)*, pages 185–191, New York, NY, 2016.

Mechthild Stoer and Frank Wagner. A simple min-cut algorithm. *Journal of the ACM*, 44(4):585–591, 1997.

Avery Wang. An industrial-strength audio search algorithm. In *Proceedings of the 4th International Conference on Music Information Retrieval (ISMIR)*, pages 7–13, Baltimore, MD, 2003.